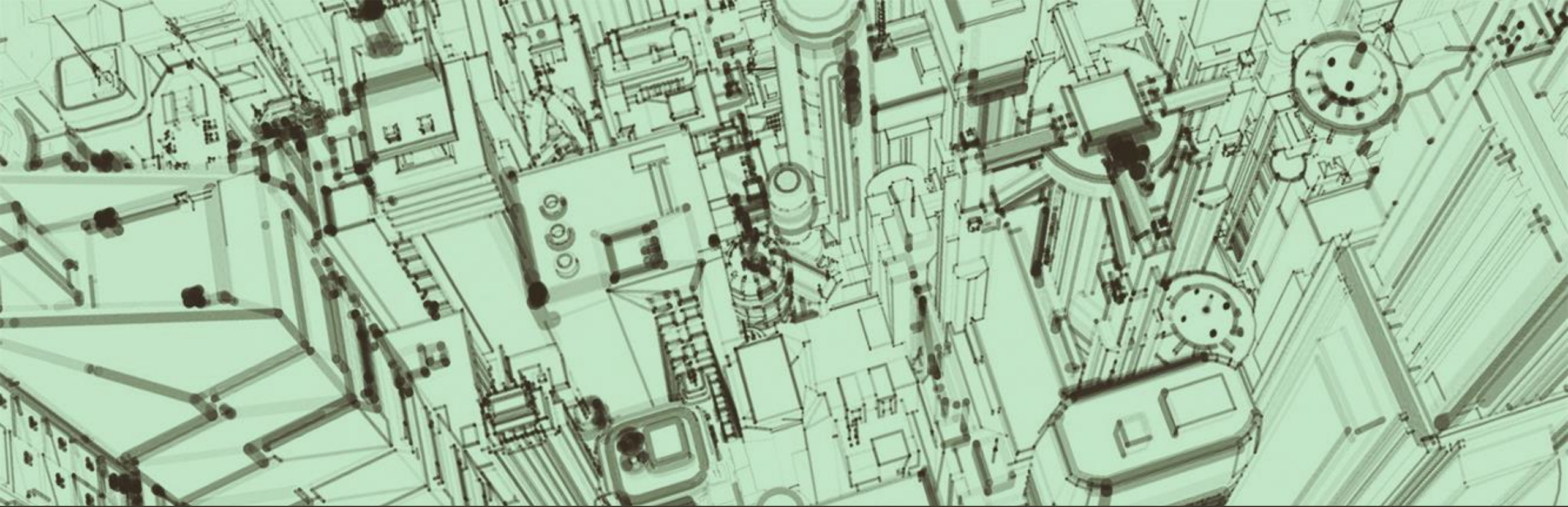




Privacy in Federated Learning

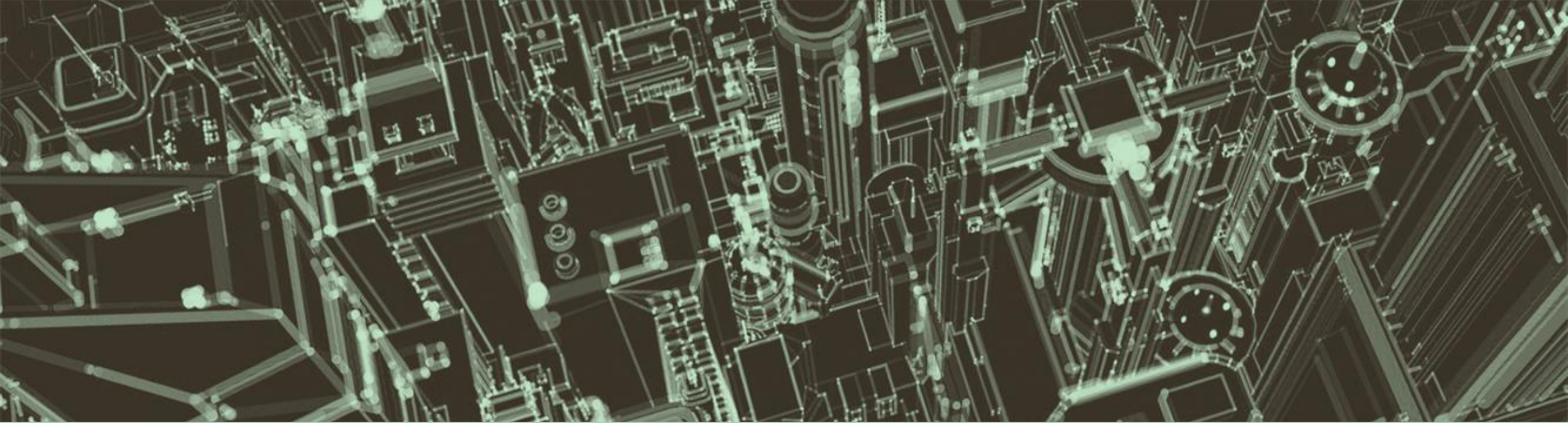
UTD Team 2 – Midterm Presentation



MEET THE TEAM

The University of Texas at Dallas – TEAM 2

- DHYANA SAI REDDY DALLI (DXD210031)
- LAKSHMI SAIEESHA BONTHALA (LXB210017)
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- NIKHIL SATISH JOSHI (NSJ210001)
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- SRINIVAS PANCHANGAM (VXP210036)
- ROLAND VARRIALE (Argonne National Laboratory (ANL)) – TECHNICAL DIRECTOR
- BRIAN RICKS –FACULTY ADVISOR



LITERATURE REVIEW

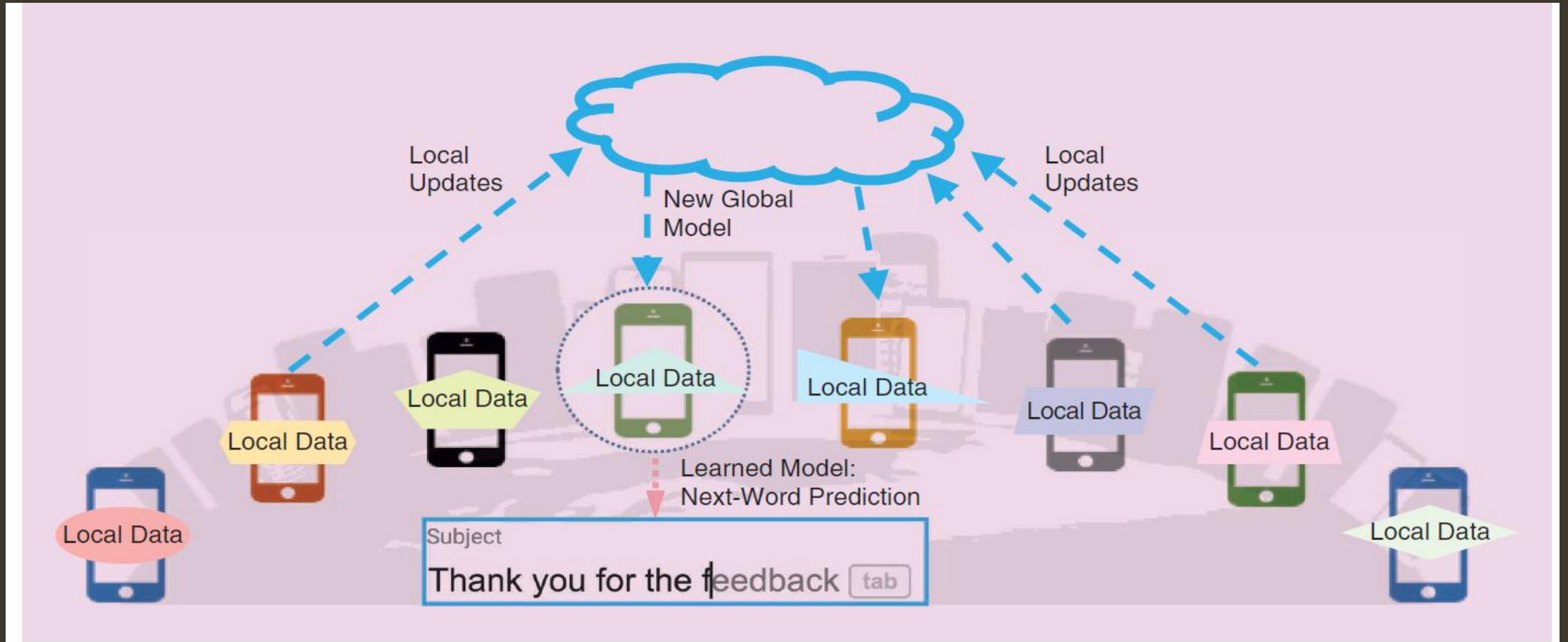
What is the project about?



INTRODUCTION TO FEDERATED LEARNING

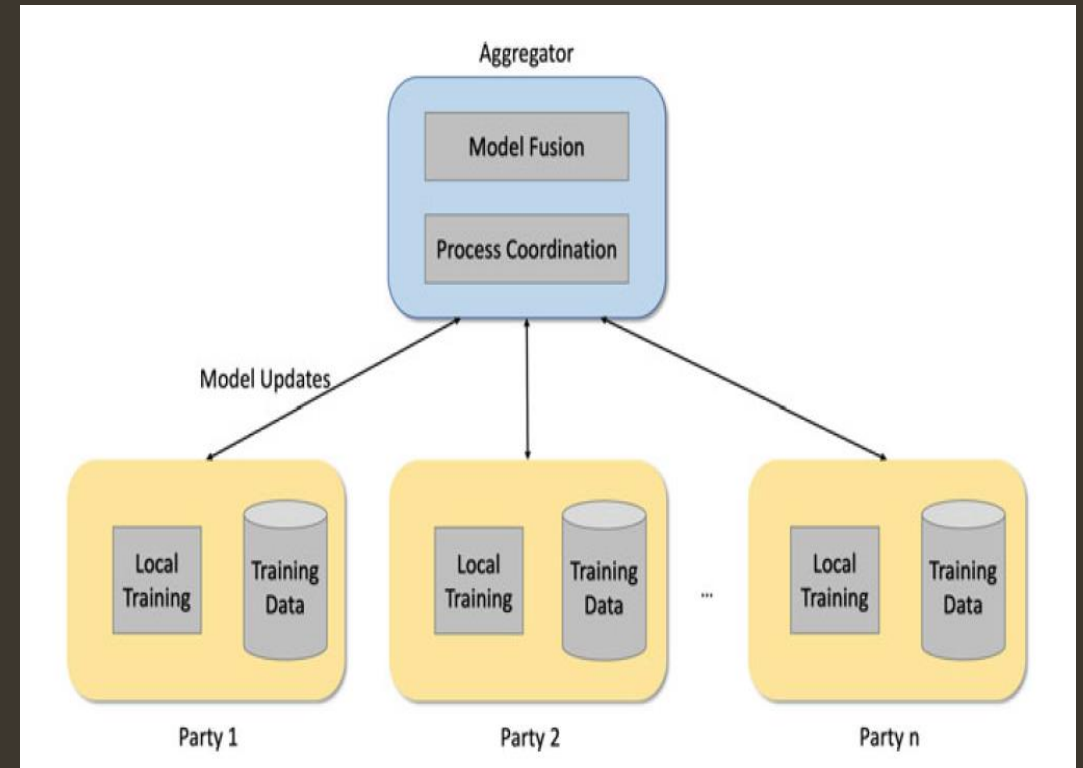
- Federated learning (FL) is a novel distributed learning paradigm that prioritizes client privacy by allowing them to retain sensitive data on their devices and only share local training parameter updates with the federated server (Aggregator).
- Federated Learning is better than traditional Machine Learning.
- Benefits of Federated Learning – Data Privacy, Reduced communication bandwidth, Increased efficiency, Adaptability

WORKING OF FEDERATED LEARNING



MOTIVATION

- Privacy
- Data Ownership
- Resource Efficiency
- Decentralization



PRIVACY LEAKAGE IN FEDERATED LEARNING

- Gradient leakages in FL may compromise the privacy of client training data.
- The malicious attacker can completely steal the training data from gradients in a few iterations
- FL protocol designs may contain vulnerabilities for:
 - 1) The (potentially malicious) server
 - 2) Any Participant
- Insider vs Outsider
- Training Phase vs Inference Phase
- Poisoning Attacks
- Inference Attacks

Training Phase vs Inference Phase

- Attacks at training phase attempt to learn, influence, or corrupt the FL model itself.
- During training phase, the attacker can run Data Poisoning Attacks or Model Poisoning Attacks.
- Attacks at inference phase do not tamper with the targeted model, but instead, either cause it to produce wrong outputs (targeted/untargeted) or collect evidence about the model characteristics.
- Inference phase attacks can be classified into:
 - White-box attacks (i.e., with full access to the FL model)
 - Black-box attacks (i.e., only able to query the FL model)

Poisoning Attacks

- Poisoning attacks can be:
 - a) Random attacks
 - b) Targeted attacks.
- Random attacks aim to reduce the accuracy of the FL model, whereas targeted attacks aim to induce the FL model to output the target label specified by the adversary.
- Poisoning attacks during the training phase can be performed on the data or on the model.
- Two poisoning attacks:
 - 1) Data poisoning (during local data collection).
 - 2) Model poisoning (during local model training process).

Data Poisoning

- Data poisoning is a cyber-attack that manipulates machine learning models by inserting false or malicious data.
- A malicious attack where an adversary intentionally injects incorrect or manipulated data into the training process of a federated learning system.
- This can cause the model to make incorrect or biased predictions, with potentially serious consequences in critical applications like Medical Diagnosis or Finance.
- Data poisoning attacks largely fall in two categories:
 - 1) Clean-Label
 - 2) Dirty-Label
- One common example of dirty-label poisoning attack is the Label-Flipping attack.
- Backdoor Poisoning.

Model Poisoning

- Model poisoning attacks aim to poison local model updates before sending them to the server or insert hidden backdoors into the global model.
- In targeted model poisoning, the adversary's objective is to cause the FL model to miss-classify a set of chosen inputs with high confidence. The misclassification is a result of adversarial manipulations of the training process.
- These poisoned updates can be generated by inserting hidden backdoors, and even a single-shot attack may be enough to introduce a backdoor into a model.
- Model poisoning attacks are much more effective than data poisoning in FL settings by analyzing a targeted model poisoning attack.

Inference Attacks

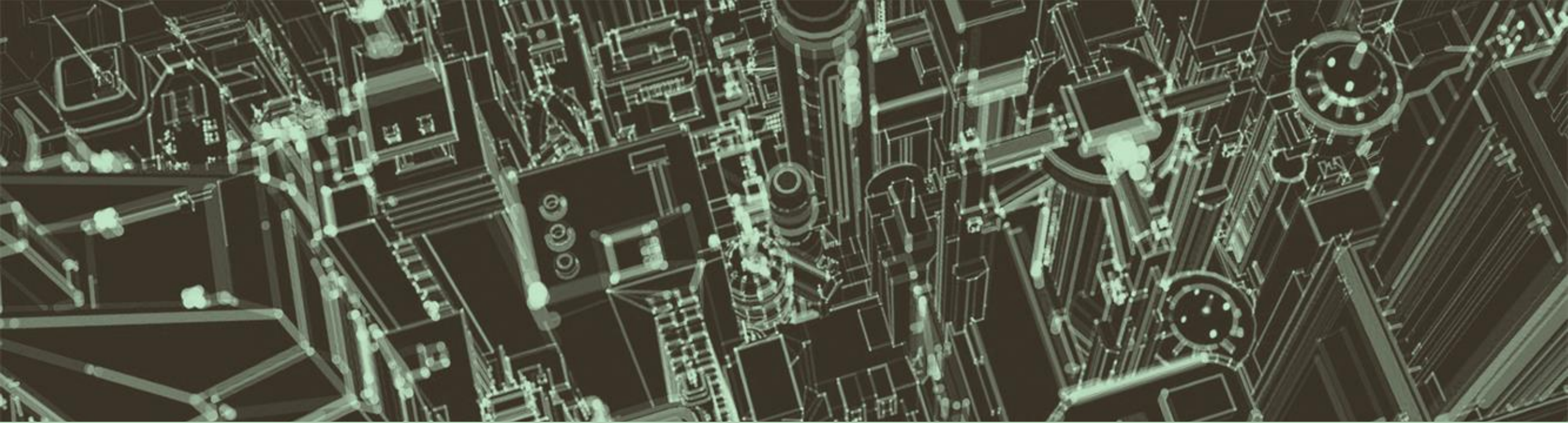
- Inference attacks are a type of attack on machine learning models that aim to extract sensitive information or infer information (characteristics) about the training data by exploiting the model's behavior.
- Inference attacks can be categorized into two main types
 - 1) Membership inference attacks
 - 2) Model inversion attacks:

Gradient Leakage Attack

- Privacy attack in federated learning where an attacker can obtain sensitive information from the gradients of the model being trained across multiple devices or servers.
- Updates to the model are generated by computing the gradients of the loss function with respect to the model parameters on each device. These gradients can be intercepted and used to infer information about the training data on each device.
- Gradient leakage can lead to privacy violations and potential legal or ethical issues, especially if the training data contains sensitive information about individuals.

Defenses Against Gradient Leakage Attack

1. Gradient Perturbation
 2. Input Encryption and Gradient Encryption
 3. Homomorphic Encryption (HE)
 4. Local Differential Privacy (LDP)
 5. Multi-Party Computation (MPC)
 6. Targeted Defense (TD)
- However, none of these defenses would work without considerable defense overheads, and none of these focuses on lightweight and guaranteed defensive mechanism against the gradient leakage attacks in FL.



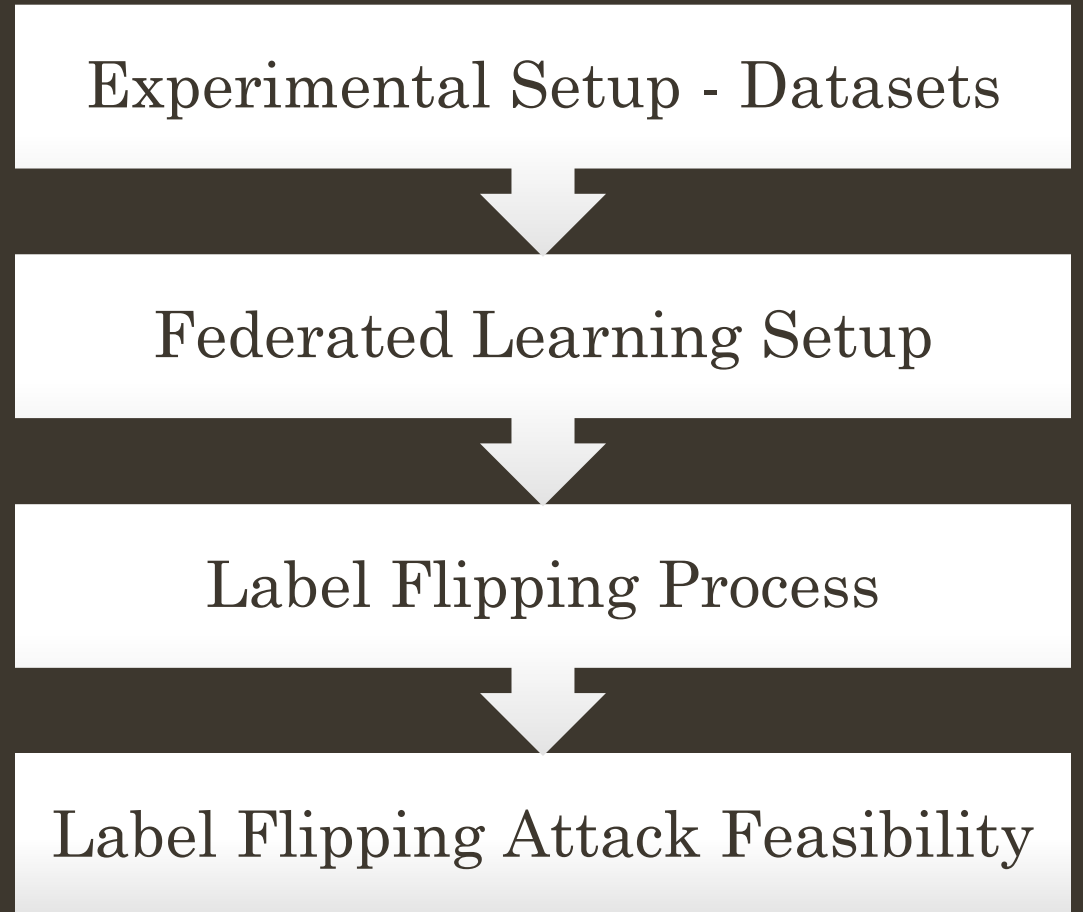
NEXT STEPS

What difference can we make?



ANALYSIS OF LABEL FLIPPING ATTACKS

- Work on various datasets and find the vulnerability percentage among them.
- FL in Python using:
 - PySyft
 - TensorFlow Federated
- APPFL
- Simulate the label flipping attack in a FL system with N participants of which $m\%$ are malicious
- Evaluation of attack feasibility and impact of malicious participant percentage on attack effectiveness.



DEFENDING AGAINST LABEL FLIPPING ATTACKS

- Algorithms to identify Malicious model Updates in FL.
- Improve the existing Algorithms for a better efficiency.

Dimensionality Reduction
(PCA)



Improve Existing Algorithms



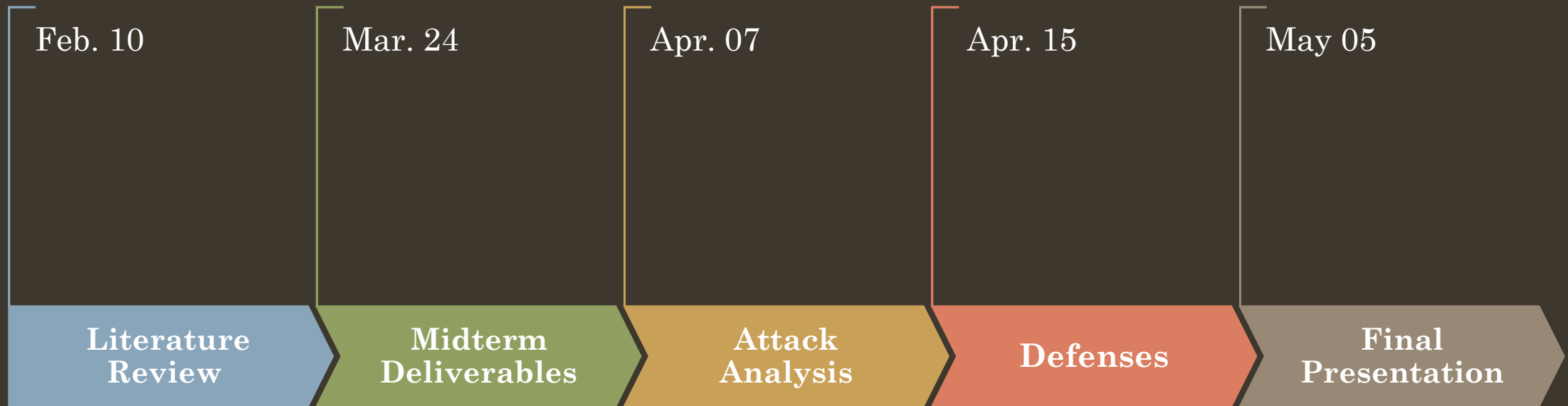
New PCA Plots

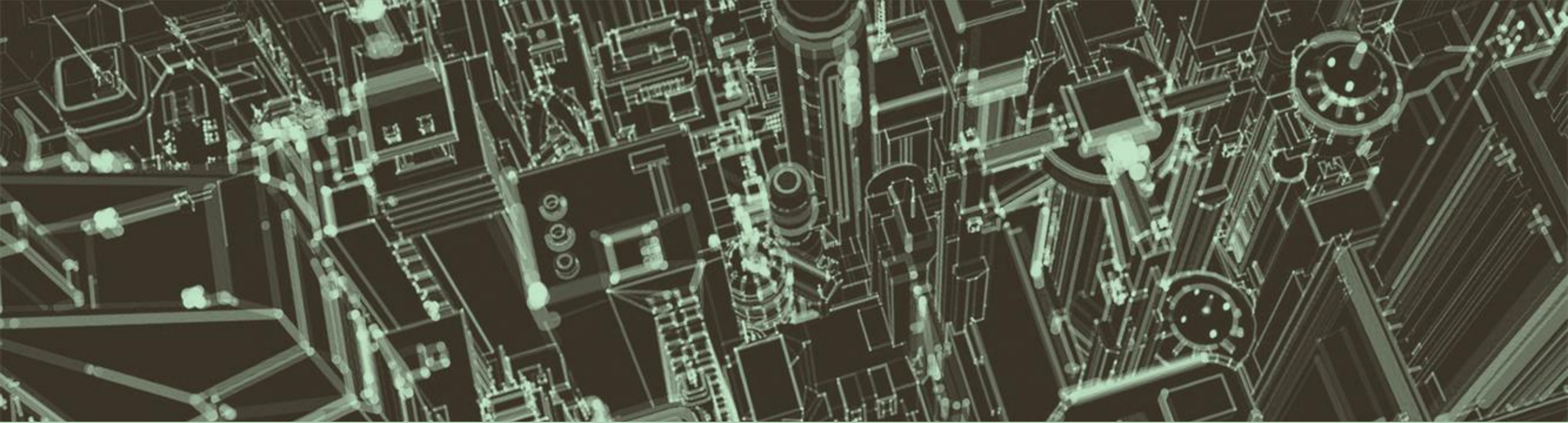
DEFENDING AGAINST LABEL FLIPPING ATTACKS

- We further plan to read more research papers and expand our knowledge on the defense mechanism that is lightweight and has less overhead.
- Once we finalize on an efficient approach, we will implement it.

Research on
Lightweight
Approaches

TIMELINE





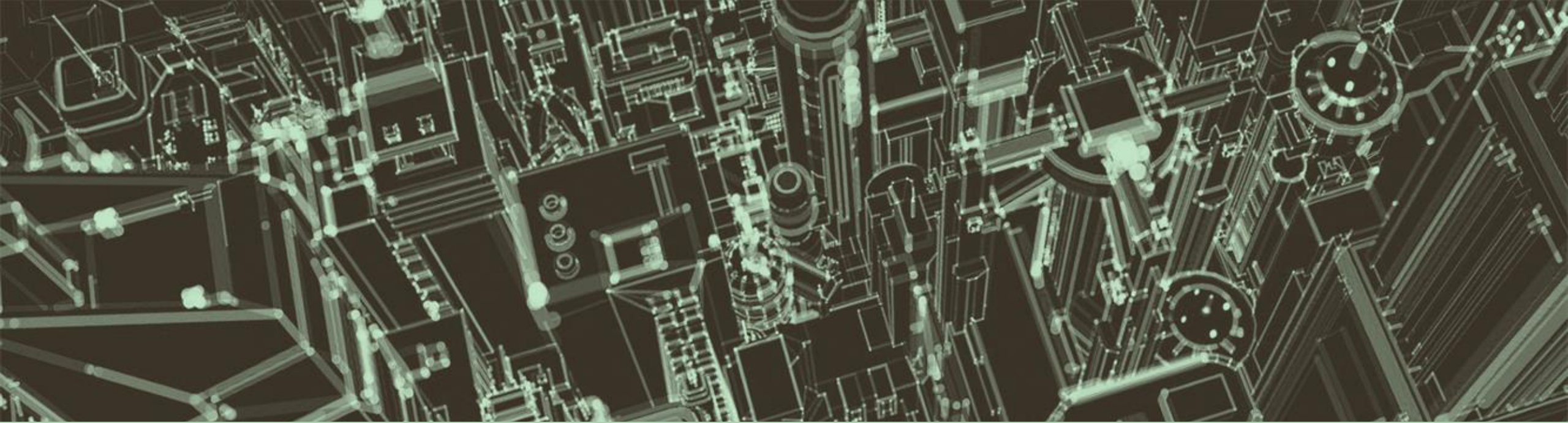
CONCLUSION

SUMMARY



SUMMARY

1. Federated learning presents unique opportunities for decentralized and collaborative machine learning, but also requires sophisticated privacy-preserving techniques to protect sensitive data
2. Techniques such as differential privacy, secure aggregation, and homomorphic encryption show promise for addressing privacy concerns in federated learning, but continued research and innovation is necessary to improve their practicality and efficiency.
3. Through ongoing collaboration and investment in new privacy-preserving techniques, we can build a future in which federated learning is a trusted and transformative tool for driving innovation while safeguarding privacy and data security.



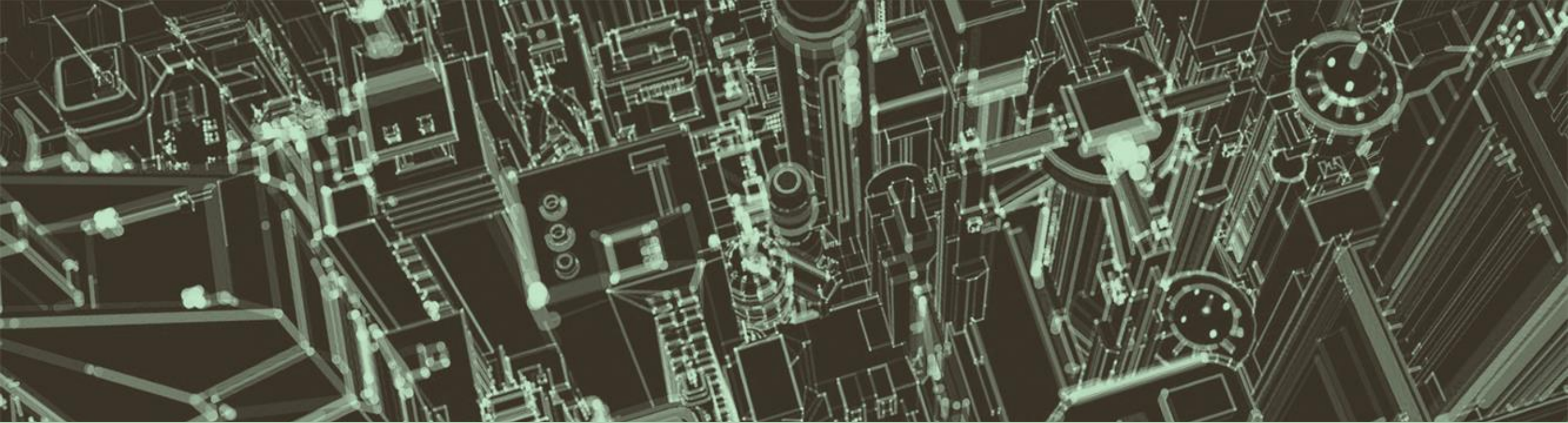
REFERENCES

All the Research papers read and referenced.



REFERENCES

1. Threats to Federated Learning: A Survey : Lingjuan Lyu, Han Yu and Qiang Yang:
<https://arxiv.org/pdf/2003.02133.pdf>
2. Data Poisoning Attacks Against Federated Learning Systems: Vale Tolpegin, Stacey Truex , Mehmet Emre Gursoy, and Ling Liu
3. Protect Privacy from Gradient Leakage Attack in Federated Learning: Junxiao Wang, Song Guo, Xin Xie, Heng Q
4. Defense Against Gradient Leakage Attacks via Learning to Obscure Data: Yuxuan Wan, Han Xu, Xiaorui Liu, Jie Ren, Wenqi Fan, Jiliang Tang
5. APPFL: Argonne Privacy-Preserving Federated Learning (APPFL) framework: <https://arxiv.org/pdf/2202.03672.pdf>
6. A Taxonomy of Attacks on Federated Learning Malhar Jere, Tyler Farnan, and Farinaz Koushanfar: <https://ieeexplore.ieee.org/document/9308910>



THANK YOU

Any Questions ?

